CN-Lec13-Looping Statement

Looping Statement

🎯 Session Objectives

- ✅ Understand conditional statements with shorthand notations.
- ✅ Understand what looping statements are
- ✅ Understand what a for loop is
- ✅ Understand what a while loop is

```python
# Conditional Statement with Core Data Structure
student_details = {
    'name' : 'Siddharth Sharan',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Jaipur',
    'state' : 'Rajasthan',
    'country' : 'India',
    'course' : 'Data Science',
    'skills' : ['Python','Numpy','Pandas','Statistics','ML','DL','NLP']
}
key_evaluate = 'course'
if key_evaluate in student_details: # in 'Membership Operator' [Boolean Returns]
    print(f"You are Searching {key_evaluate}: {student_details[key_evaluate]}")
else:
    print("No Such Key Exist.")
```

```
You are Searching course: Data Science
```

```python
# Conditional Statement with Core Data Structure
student_details = {
    'name' : 'Siddharth Sharan',
    'age' : 29,
    'gender' : 'Male',
    'city' : 'Jaipur',
    'state' : 'Rajasthan',
    'country' : 'India',
    'course' : 'Data Science',
    'skills' : ['Python','Numpy','Pandas','Statistics','ML','DL','NLP']
}
key_evaluate = 'skills'
if key_evaluate in student_details: # in 'Membership Operator' [Boolean Returns]
    print(f"You are Searching {key_evaluate}: {student_details[key_evaluate]}")
else:
    print("No Such Key Exist.")
```

```
You are Searching skills: ['Python', 'Numpy', 'Pandas', 'Statistics', 'ML', 'DL', 'NLP']
```

```python
# Conditional Statement with Core Data Structure
key_evaluate = 'email'
if key_evaluate in student_details: # in 'Membership Operator' [Boolean Returns]
    print(f"You are Searching {key_evaluate}: {student_details[key_evaluate]}")
else:
    print("No Such Key Exist.")
```

```
No Such Key Exist.
```

```python
if condition1:
    # Block if condition1 true
elif condition2:
    # Block if condition2 true
elif condition3:
    # Block if condition3 true
...
else:
    # Block if all conditions are false
```

```python
num = 900
if num == 1000:
    print(f"{num} is equal to 1000.")
elif num >= 700:
    print(f"{num} is greater than or equal to 700.")
elif num >= 500:
    print(f"{num} is greater than or equal to 500.")
else:
    print(f"{num} is less than 500.")
```

```
900 is greater than or equal to 700.
```

```python
# Find the max marks by comparison
marks1 = int(input("Enter the Marks1: "))
marks2 = int(input("Enter the Marks2: "))
marks3 = int(input("Enter the Marks3: "))

if marks1 > marks2 and marks1 > marks3: # marks1 is winner
    print(f"{marks1} is greater than both {marks2} & {marks3}")
elif marks2 > marks1 and marks2 > marks3: # marks2 is winner
    print(f"{marks2} is greater than both {marks1} & {marks3}")
else: # marks3 is winner
    print(f"{marks3} is greater than both {marks1} & {marks2}")
```

```
Enter the Marks1:  91
Enter the Marks2:  82
Enter the Marks3:  73
91 is greater than both 82 & 73
```

```python
# Find the max marks by comparison
marks1 = int(input("Enter the Marks1: "))
marks2 = int(input("Enter the Marks2: "))
marks3 = int(input("Enter the Marks3: "))

if marks1 > marks2 and marks1 > marks3: # marks1 is winner
    print(f"{marks1} is greater than both {marks2} & {marks3}")
elif marks2 > marks1 and marks2 > marks3: # marks2 is winner
    print(f"{marks2} is greater than both {marks1} & {marks3}")
else: # marks3 is winner
    print(f"{marks3} is greater than both {marks1} & {marks2}")
```

```
Enter the Marks1:  81
Enter the Marks2:  92
Enter the Marks3:  79
92 is greater than both 81 & 79
```

```python
# Find the max marks by comparison
marks1 = int(input("Enter the Marks1: "))
marks2 = int(input("Enter the Marks2: "))
marks3 = int(input("Enter the Marks3: "))

if marks1 > marks2 and marks1 > marks3: # marks1 is winner
    print(f"{marks1} is greater than both {marks2} & {marks3}")
elif marks2 > marks1 and marks2 > marks3: # marks2 is winner
    print(f"{marks2} is greater than both {marks1} & {marks3}")
else: # marks3 is winner
    print(f"{marks3} is greater than both {marks1} & {marks2}")
```

```
Enter the Marks1:  77
Enter the Marks2:  81
Enter the Marks3:  95
95 is greater than both 77 & 81
```

Shorthand Conditional Statements :

if : if condition: statement

if-else : result = value1 if condition else value2

if-elif-else : result = (value1 if condition1 else value2) if condition2 else value3

result = (value1 if condition1 else value2) if condition2 else value3

Syntax : If

```
If condition:
    # code block
```

Syntax : If-else

```
if condition:
    # Code block to execute if the condition is true
else:
    # Code block to execute if the condition is false
```

```
# Shorthand Conditional Statement
marks = int(input("Enter the Valid Marks: "))
# if condition: statement
if marks>=90 : print("Excellent")
```

```
Enter the Valid Marks:  91
Excellent
```

```
# Shorthand Conditional Statement
marks = int(input("Enter the Valid Marks: "))
# if condition: statement
if marks>=90 : print("Excellent")
```

```
Enter the Valid Marks:  77
```

```
# if-else:
# result = value1 if condition else value2
marks = int(input("Enter the Valid Marks: "))
result = "pass" if marks>=33 else "fail"
print(result)
```

```
Enter the Valid Marks:  55
pass
```

```
# if-else:
# result = value1 if condition else value2
marks = int(input("Enter the Valid Marks: "))
result = "pass" if marks>=33 else "fail"
print(result)
```

```
Enter the Valid Marks:  32
fail
```

```
# if-else:
# result = value1 if condition else value2
marks = int(input("Enter the Valid Marks: "))
print("pass") if marks>=33 else print("fail")
```

```
Enter the Valid Marks:  92
pass
```

```
num = 900
if num == 1000:
    print(f"{num} is equal to 1000.")
elif num >= 500:
    print(f"{num} is greater than or equal to 500.")
else:
    print(f"{num} is less than 500.")
```

```
# result = (value1 if condition1 else value2) if condition2 else value3
result = (print(f"{num}= 1000.") if num == 1000 else print(f"{num}>=500.")) if num >= 500 else print(f"{num}<50
```

```
900>=500.
```

```
# Find the max marks by comparison
marks1 = int(input("Enter the Marks1: "))
marks2 = int(input("Enter the Marks2: "))
marks3 = int(input("Enter the Marks3: "))

# if marks1 > marks2 and marks1 > marks3: # marks1 is winner
#     print(f"{marks1} is greater than both {marks2} & {marks3}")
# elif marks2 > marks1 and marks2 > marks3: # marks2 is winner
#     print(f"{marks2} is greater than both {marks1} & {marks3}")
# else: # marks3 is winner
#     print(f"{marks3} is greater than both {marks1} & {marks2}")

result = ("Marks1 is Winner" if (marks1>marks2 and marks1>marks3) else "Marks2 is Winner") if (marks2>marks1 ar
print(result)
```

```
Enter the Marks1:  77
Enter the Marks2:  81
Enter the Marks3:  95
Marks3 is Winner
```


```
# result = (value1 if condition1 else value2) if condition2 else value3
result = (print(f"{num}= 1000.") if num == 1000 else print(f"{num}>=500."))
        if num >= 500 else print(f"{num}<500.")
```

```
result = ("Marks1 is Winner" if (marks1>marks2 and marks1>marks3) else
        "Marks2 is Winner") if (marks2>marks1 and marks2>marks3)
        else "Marks3 is Winner"
print(result)
```

```
Enter the Marks1:  77
Enter the Marks2:  81
Enter the Marks3:  95
Marks3 is Winner
```


```
# ASCII Comparision
val1 = 'Apple'
val2 = 'apple'
result = (val1 if val1>val2 else val2)
print(result)
```

```
apple
```

```
# ASCII Comparision
val1 = 100
val2 = 100
result = (val1 if val1>val2 else val2)
print(result) # val2
```

```
100
```

```
# ASCII Comparision
height1 = 181
height2 = 192
result = (height1 if height1>height2 else height2)
print(result) # height2
```

```
192
```

```python
# Odd [%2 == 1] [%2 != 0] / Even [%2 == 0]
val = int(input("Enter the Value: "))
result = ("Even" if val%2==0 else "Odd")
print(result)
```

```
Enter the Value:  99
Odd
```

```python
val = int(input("Enter the Value: "))
result = ("Even" if val%2==0 else "Odd")
print(result)
```

```
Enter the Value:  82
Even
```

```python
val = int(input("Enter the Value: "))
if val%2 ==0:
    result = "Even"
else:
    result = "Odd"
print(result)
```

```
Enter the Value:  91
Odd
```

```python
val = int(input("Enter the Value: "))
if val%2 !=0:
    result = "Odd"
else:
    result = "Even"
print(result)
```

```
Enter the Value:  77
Odd
```

```python
val = int(input("Enter the Value: "))
if val%2 == 1:
    result = "Odd"
else:
    result = "Even"
print(result)
```

```
Enter the Value:  95
Odd
```

```python
# Grading System...
marks = int(input("Enter the Marks: "))
# result = (value1 if condition1 else value2) if condition2 else value3
result = ("A" if marks>=90 else "B") if marks >=70 else "C"
print(result)
```

```
Enter the Marks:  81
B
```

```python
marks = int(input("Enter the Marks: "))
# result = (value1 if condition1 else value2) if condition2 else value3
result = ("A" if marks>=90 else "B") if marks >=70 else "C"
print(result)
```

```
Enter the Marks:  92
A
```

```python
# Grading System...
marks = int(input("Enter the Marks: "))
grade = ""
if marks>=90:
    grade = "A+"
elif marks>=80:
    grade = "A"
elif marks>=70:
    grade = "B+"
elif marks>=60:
    grade = "B"
elif marks>=50:
    grade = "C"
elif marks>=40:
    grade = "D"
else:
    grade = "E"

print(grade)
```

```
Enter the Marks:  77
B+
```

```python
marks = 55
result = "A" if marks>=90 else "B" if marks >=70 else "C" if marks >=50 else "D"
print(result)
```

```
C
```

```python
marks = 72
result = "A" if marks>=90 else "B" if marks >=70 else "C" if marks >=50 else "D" if marks >= 30 else "E"
print(result)
```

```
B
```

Looping Statement

```python
# Looping Statement
statement = "I love Python Language 😍"
for i in range(0,100): # [0,1,2,3...99]
    print(statement)
```

```
I love Python Language 😍
I love Python Language 😍
I love Python Language 😍
I love Python Language 😍
I love Python Language 😍
I love Python Language 😍
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I love Python Language 😍
I love Python Language 😍
```

```python
# Looping is What? An Iterator iterating on Iterables.
# Even and Odd Number
_list = [11,22,100,121,216,719,912,1000,33,55,77,81,90,50]
for val in _list:
    if val % 2 == 1: # Odd
        print(f"{val} is an odd value.")
    else:
        print(f"{val} is an even value.")
print("Loop is Ended")
```

Memory

val

_list = [11,22....50]

val = ~~11~~ ~~22~~ ~~100~~ ~~121~~ ... ~~50~~

None

```
11 is an odd value.
22 is an even value.
100 is an even value.
121 is an odd value.
216 is an even value.
719 is an odd value.
912 is an even value.
1000 is an even value.
33 is an odd value.
55 is an odd value.
77 is an odd value.
81 is an odd value.
90 is an even value.
50 is an even value.
Loop is Ended
```

```python
odd_list = []
even_list = []
_list = [11,22,100,121,216,719,912,1000,33,55,77,81,90,50]
for val in _list:
    if val % 2 == 1: # Odd
        odd_list.append(val)
        print(f"{val} -> Appending to the odd_list : {odd_list}")
    else:
        even_list.append(val)
        print(f"{val} -> Appending to the even_list : {even_list}")
```

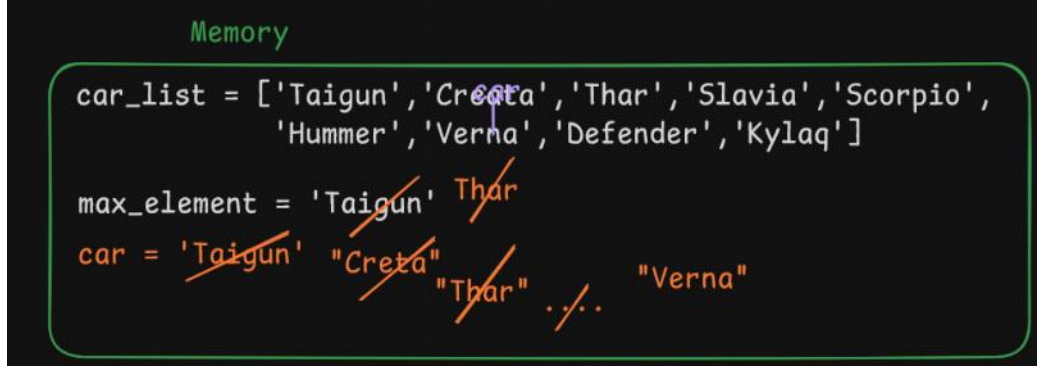
```
11 -> Appending to the odd_list : [11]
22 -> Appending to the even_list : [22]
100 -> Appending to the even_list : [22, 100]
121 -> Appending to the odd_list : [11, 121]
216 -> Appending to the even_list : [22, 100, 216]
719 -> Appending to the odd_list : [11, 121, 719]
912 -> Appending to the even_list : [22, 100, 216, 912]
1000 -> Appending to the even_list : [22, 100, 216, 912, 1000]
33 -> Appending to the odd_list : [11, 121, 719, 33]
55 -> Appending to the odd_list : [11, 121, 719, 33, 55]
77 -> Appending to the odd_list : [11, 121, 719, 33, 55, 77]
81 -> Appending to the odd_list : [11, 121, 719, 33, 55, 77, 81]
90 -> Appending to the even_list : [22, 100, 216, 912, 1000, 90]
50 -> Appending to the even_list : [22, 100, 216, 912, 1000, 90, 50]
```

```python
odd_list = []
even_list = []
_list = [11,22,100,121,216,719,912,1000,33,55,77,81,90,50]
for val in _list:
    if val % 2 == 1: # Odd
        odd_list.append(val)
    else:
        even_list.append(val)
print(f"Odd_List : {odd_list}")
print(f"Even_List : {even_list}")
```

```
Odd_List : [11, 121, 719, 33, 55, 77, 81]
Even_List : [22, 100, 216, 912, 1000, 90, 50]
```

```python
# min/max -> Loops [Categorical Records -> Alphabetical Order -> ASCII Value]
car_list = ['Taigun','Creata','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq']
max_element = car_list[0] # 'Taigun'
for car in car_list:
    if car > max_element:
        max_element = car
print(max_element) # 'Verna'
```

```
Verna
```




```python
# min/max -> Loops [Categorical Records -> Alphabetical Order -> ASCII Value]
car_list = ['XUV 7XO','Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq']
max_element = car_list[0] # XUV 7XO
for car in car_list:
    if car > max_element:
        max_element = car
print(max_element) # XUV 7XO
```

```
XUV 7XO
```

```python
# min/max -> Loops [Categorical Records -> Alphabetical Order -> ASCII Value]
car_list = ['Taigun','Creta','Thar','Slavia','Scorpio','Hummer','Verna','Defender','Kylaq','Astor']
min_element = car_list[0] #'Taigun'
for car in car_list:
    if car < min_element:
        min_element = car
print(min_element) # 'Astor'
```

```
Astor
```